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Supplementary Information for

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**W. Ma, P. Alonso-González and S. Li contributed equally to this work.

The Supplementary Information consists of 11 sections with complementary details of the results obtained in the main manuscript as described below:

- **Section S.1.** Infrared reflectivity of α-MoO_3 crystals characterization by Fourier transform infrared (FTIR) spectroscopy.
- **Section S.2.** Anisotropic structure of α-MoO_3 crystals: characterization by scanning electron microscopy (SEM), Raman spectroscopy, and high-resolution transmission electron microscopy (HRTEM). A discussion of the relationship between the structural anisotropy of α-MoO_3 crystals and their optical properties is given.
- **Section S.3**. Sign of the α-MoO_3 phonon polaritons (PhPs) phase velocities: experimental extraction by scattering-type near-field optical microscopy (s-SNOM) of the sign of α-MoO_3 phonon polaritons phase velocities.
- **Section S.4.** Group velocities of α-MoO_3 PhPs: derivation of the group velocities of α-MoO_3 PhPs from their experimental optical dispersions.
- **Section S.5.** Edge- and tip-launched PhPs in α-MoO_3: analysis of the different periodicities by s-SNOM of edge- and tip-launched PhPs in α-MoO_3
- **Section S.6.** s-SNOM polariton interferometry in α-MoO_3: Interpretation of s-SNOM polariton interferometry images in a hyperbolic material.
- **Section S.7.** Derivation of the polariton dispersion along a thin anisotropic sheet: dispersion relation, thickness dependence of the polariton wavevector, and numerical simulations of s-SNOM polariton interferometry images.
- **Section S.8.** Notes for the fabrication of the α-MoO_3 disk.
- **Section S.9.** Lifetimes of PhPs in α-MoO_3: calculation of the lifetimes from s-SNOM profiles along the [100] α-MoO_3 direction.
- **Section S.10.** Lifetime dependence of α-MoO_3 PhPs on the flake thickness.
- **Section S.11.** Group velocity and lifetime of PhPs in an ultra-low-loss α-MoO_3 flake.

<u>S.1. Infrared reflectivity of α-MoO_3 crystals.</u>

To characterize the far-field optical response of α-MoO_3, we performed polarization-resolved FTIR spectroscopy on an individual α-MoO_3 flake. Figure S1a shows the polarization-dependent reflection spectra of an α-MoO_3 flake on a SiO_2/Si substrate. For all incident polarization directions, excepting at 0º degrees (see inset), we observe a broad main peak in the reflectivity spectrum with its maximum value dependent on the polarization angle, and being largest at 90º. This angle-dependent reflectivity supports the assumption of anisotropic optical response in α-MoO_3 based on its crystal structure (Fig. 1 in the main text). The strong IR reflection between about 800 cm^{-1} and 1000 cm^{-1} is assigned to two IR "Reststrahlen bands" (RBs), originated from asymmetric bond stretching modes of the lattice[1,2] (see section S.2).

We performed micro-FTIR transmission measurements (see Fig. S1b and S1c) and found that the optical response in these two RBs presents a strong dependence on the incident polarization. The low-refractive index material BaF_2 was used as the substrate as it does not exhibit phonon resonances in the spectral range considered in our work. For L-RB (Fig. S1b), the highest optical transmission is along the [100] direction, whereas it remains nearly indiscernible along the orthogonal [001] direction. This suggests that the optical responses along [001] and [100] crystalline directions are very divergent, and the giant in-plane difference is correlated with the strongly in-plane anisotropic permittivity (the in-plane permittivity has opposite signs as we discussed in the main text). This also explains the fact that the optical phonons along [100] direction dominate the behavior of in-plane hyperbolic polaritons in L-RB. On the other hand, for the U-RB shown in Fig. S1c, the optical phonons along [100] and [001] directions exhibit a different degree of transmission and a prominent frequency shift, which indicates that the anisotropy is smaller than in the L-RB. It is noted that the in-plane permittivities in the U-RB along [100] and [001] crystalline directions are different but with the same sign. Consequently, α-MoO_3 supports in-plane elliptical polaritons in the U-RB.

Briefly, the directional optical phonons in the plane render the directional propagating PhPs, and the divergence of in-plane optical phonons in a particular RB may contribute to the degree of in-plane anisotropy (in-plane elliptical or hyperbolic anisotropy).

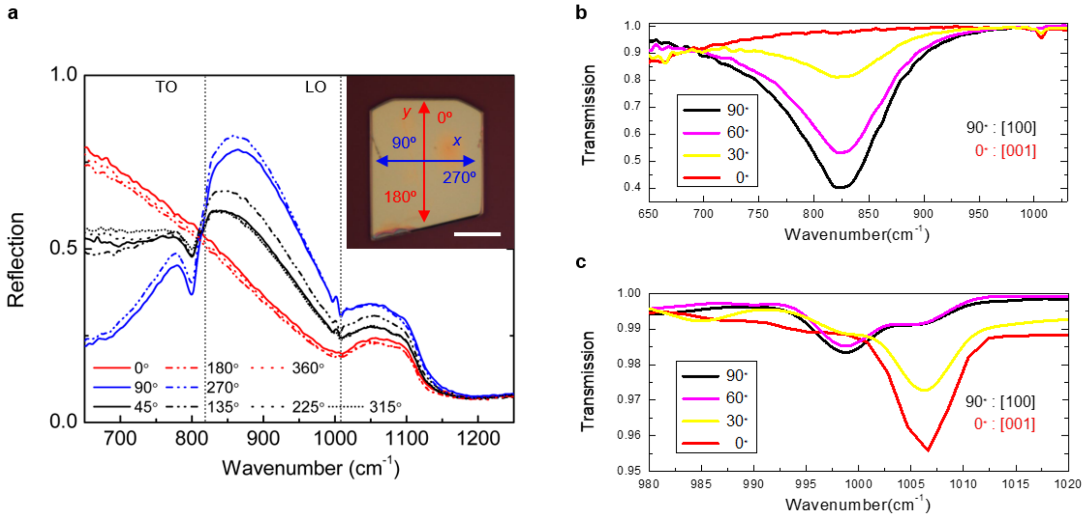

Figure S1. IR reflection and transmission of α-MoO_3 flakes. **a.** FTIR reflection spectra of an α-MoO_3 flake for various polarization angles of the incident light. Inset: optical image of the α-MoO_3 crystal. The arrows indicate light polarization directions ("x" and "y" axes are the [100] and [001] crystalline directions, respectively). Scale bar: 5 μm. **b.** FTIR transmission spectra of an α-MoO_3 flake on a BaF_2 substrate for various polarization angles of the incident light. **c.** Zoom-in transmission spectra spanning the U-RB in Fig. S1b.

S. 2. Anisotropic structure of α-MoO_3 crystals.

S.2.1 Structural characterization.

A representative SEM image of the α-MoO_3 flakes is shown in Fig. S2a. Due to their anisotropic crystal structure, the α-MoO_3 crystals typically appear to be a rectangular-like shape in which the long-axis corresponds to the [001] direction. Previous works have reported that ribbon-type α-MoO_3 flakes oriented along the [001] direction are favored upon exfoliation because the energy release along the [001] direction is much greater than that along the [100] direction.[3,4] We also further verified this favored orientation by high-resolution transmission electron microscopy (HRTEM) (see Fig. S3). Thus, the crystallographic direction can be easily determined by either optical microscopy (Fig. 1a) or electron microscopy (Fig. S2a).

The Raman spectrum of an α-MoO_3 flake is shown in Fig. S2b. It reveals the characteristic peaks of an orthorhombic α-MoO_3 phase. The peak at 666 cm^{-1} is attributed to a triple-coordinated oxygen stretching mode, the peak at 820 cm^{-1} is attributed to a double-coordinated oxygen stretching mode (Fig. S4d), and the peak at 996 cm^{-1} is attributed to a terminal oxygen stretching mode (Fig. S4e)[5]. Those two peaks at 820 cm^{-1} (Fig. S2c) and 996 cm^{-1} (Fig. S2d) are assigned to the same lattice vibrations as those originating the L- and U-RBs in α-MoO_3, respectively.[1] As the Raman peaks are extremely sharp (full-width at half-maximum (FWHM) = 8.1 and 2.6 cm^{-1}, respectively), we can thus expect long PhPs lifetimes in α-MoO_3. Note that a relation between the PhPs lifetime and the Raman peak linewidth has been recently verified for ultralow-loss polaritons in isotopically pure boron nitride[6], which indicates that the PhPs lifetime is not expected to depend on the sample thickness (see section S.10).

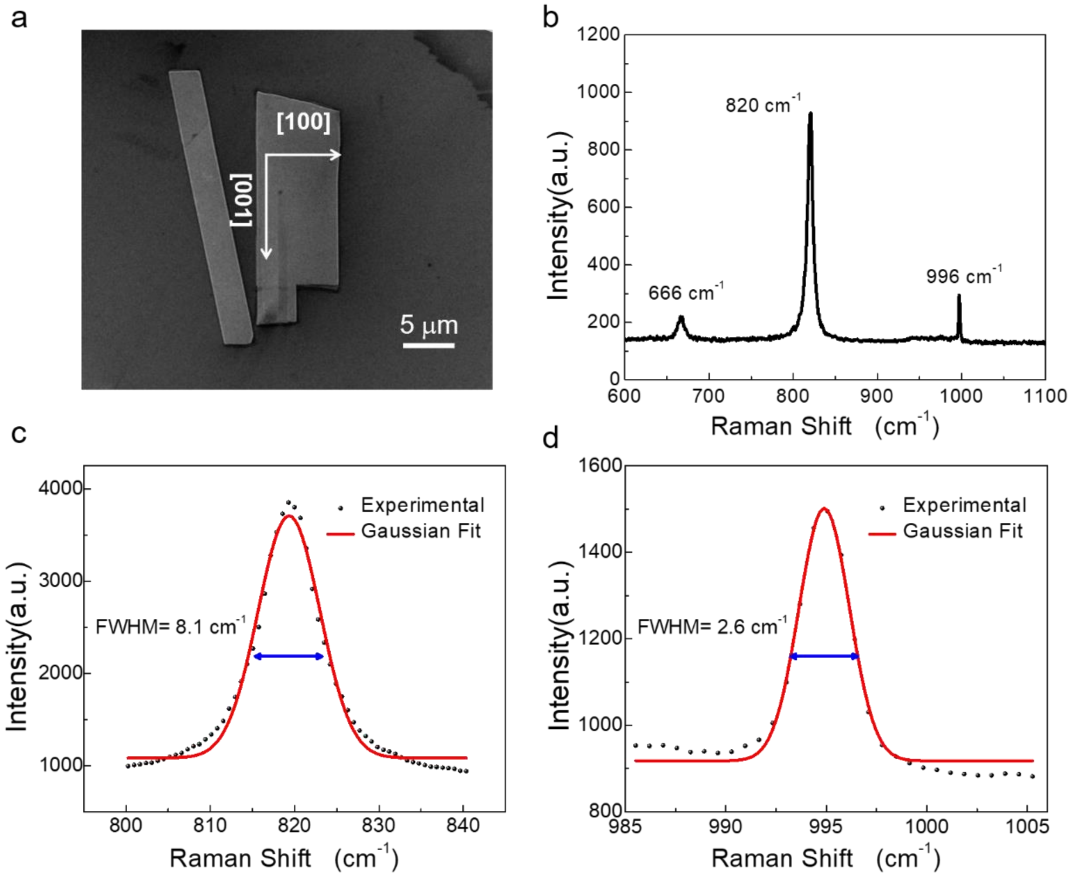

Figure S2. a. Representative SEM image of α-MoO_3. The arrows indicate the crystal directions. **b.** Raman spectrum taken from the crystal in the middle. **c,d.** Zoom-in spectra showing the Raman peaks at 820 cm^{-1} and 996 cm^{-1} as well as Gaussian fitting.

Fig. S3a, and S3b show HRTEM images and the corresponding selective-area electron diffraction (SAED) patterns of an α-MoO_3 flake. They clearly indicate the characteristic orthorhombic structure of α-MoO_3. Two sets of parallel fringes appear with a spacing of 0.37 nm and 0.39 nm for the (001) and (100) facet, respectively. The corresponding SAED pattern is recorded with the incident electron beam perpendicular to the nanoflake, revealing the [001] and [100] crystallographic directions of the α-MoO_3 flake[5]. When a similar flake was measured by s-SNOM polariton interferometry (Fig. S3c, and S3d), we unambiguously assign the [100] direction of the crystal to the direction of preferential propagation of α-MoO_3 PhPs in the L-RB (ω= 902 cm^{-1}).

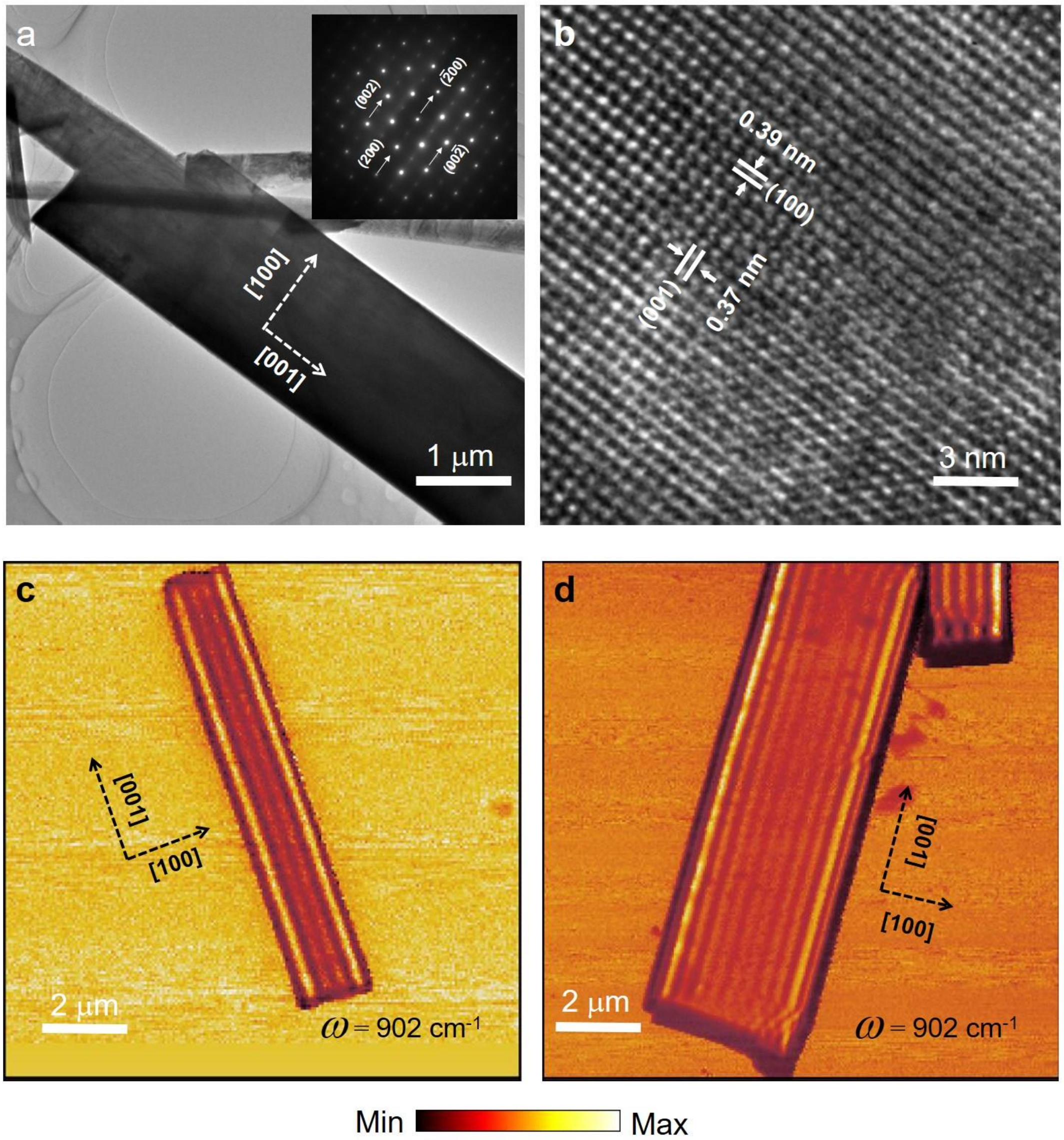

Figure S3. a. TEM image of an α-MoO_3 crystal. Inset: SAED pattern of the α-MoO_3 crystal. The (002) and (200) facets are indicated by the arrows. **b.** HRTEM image of the α-MoO_3 crystal. **c, d.** s-SNOM near-field amplitude images of two α-MoO_3 crystals taken in the L-RB (ω= 902 cm^{-1}).

S.2.2. Relationship between the structural and optical anisotropy in α-MoO_3 .

Large optical anisotropy is generally found in structurally anisotropic crystals, for example, strong birefringence or dichroism has been observed in biaxial crystals with orthorhombic structure[7,8]. MoO_3 crystal has a very interesting orthorhombic layered structure. Most importantly, it has a strong in-plane structure anisotropy as the difference of the spacing between (100) facets and (001) facets is as large as 7.2 %.
The optical anisotropy in a crystal is quantified by the differences between the real parts of complex refractive indices as birefringence (Δn) and between the imaginary parts of the indices as dichroism ($\Delta \kappa$). The refractive index of a homogeneous medium can be described by the polarizability of its constituent atoms, ions or molecules by following the Clausius–Mossotti relation[9]. Intuitively, a large optical anisotropy can be achieved by tuning the anisotropy of the polarizability tensor, which requires controlling the nature and distribution of the constituent elements[10]. At infrared frequencies, the largest contribution to the refractive index often arises from the electronic polarizability. Electronic polarizability is essentially due to the dipoles forming in individual ions via the distortion of the electronic cloud from the nucleus.

α-MoO_3 consists of double-layers of linked distorted MoO_6 octahedra. In each double-layer, MoO_6 forms corner sharing chains along the [001] direction and edge sharing chains along the [100] direction as shown in Fig. 1a and Fig. S4a-c. There are totally three O atoms differently linked with one Mo atom, of which O_2-Mo is along the [001] direction and O_3-Mo along the [100] direction (see Fig. S4d,e). The O^{2-} will possess different electronic polarizabilities in different environments (0.5 - 3.2 Å^3) while the electronic polarizabilities of Mo^{6+} is 0.048 Å^3.[9] Largely different electronic polarizabilities can be expected for in-plane O_2 (along [001]) and O_3 (along [100]), which could result in strong in-plane optical anisotropy. The precise calculation of electronic polarizabilities of O_2 or O_3 in different circumstances in α-MoO_3 is out of the scope of the current study but worthy of further investigation.

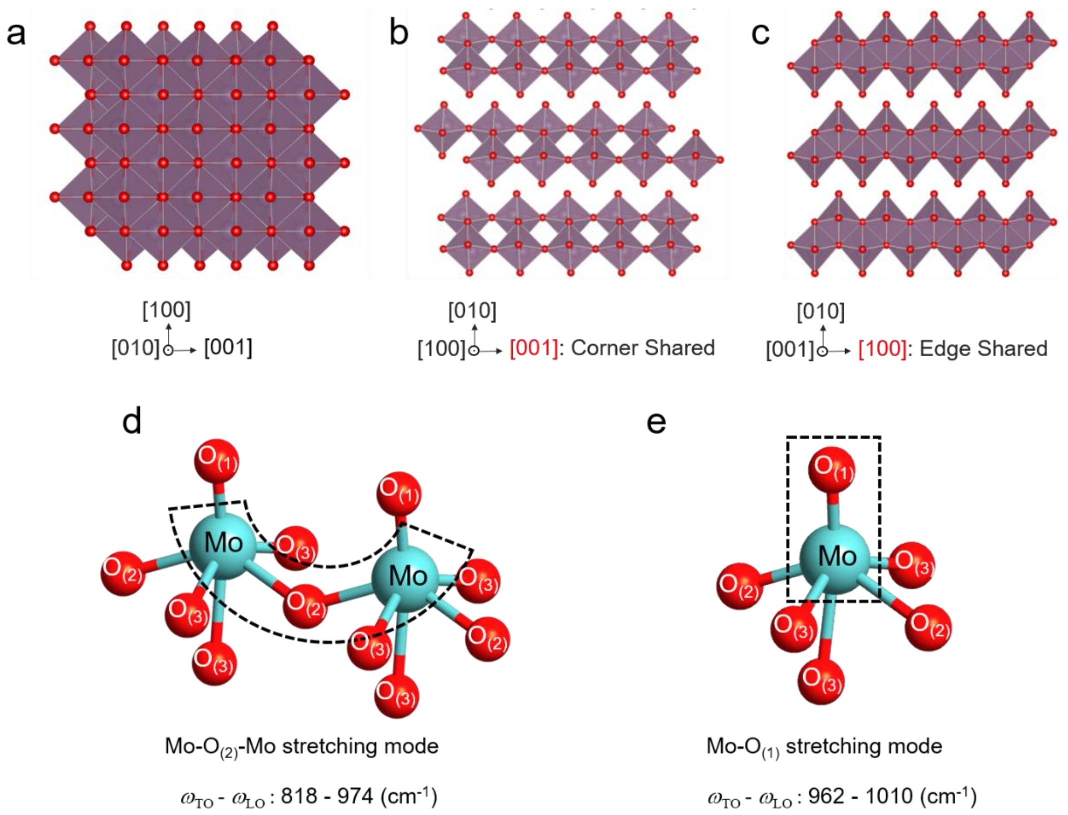

Figure S4. a,b,c. Polyhedral representation of the orthorhombic phase of α-MoO_3 viewed from [010], [100],[001] direction. **d,e.** Anisotropic bond stretching modes in α-MoO_3. **d.** Mo-$O_{(2)}$-Mo bond stretching. **e.** Mo-$O_{(1)}$ bond stretching. Due to the inequivalent sites of the oxygen atoms ($O_{(1)}$,$O_{(2)}$,$O_{(3)}$), these bond stretching modes induce anisotropic phonon vibrations in α-MoO_3.

S.3. Sign of the α-MoO_3 PhPs phase velocities.

To extract the sign of the α-MoO_3 PhPs phase velocities in the L- and U-RBs, we plot the line profiles of the imaginary and real parts of the complex-valued s-SNOM signal σ_4 (at representative frequencies of both RBs: ω= 900 cm^{-1}, and ω= 990 cm^{-1}, for the L- and the U-RB, respectively) (Fig. S5). By doing so, we observe an anticlockwise rotating spiral in the L-RB region (red lines), while it is clockwise in the U-RB region (blue lines), revealing the positive and negative phase velocities in these regions, respectively[11]. Note that the center of rotation of the spiral shows a lateral displacement in the plots, which is explained by the fact that the represented profiles contain contributions of both tip- and edge-launched PhPs (see section S.5).
By performing a 2D-FFT filtering in the experimental s-SNOM profiles, we can isolate the near-field contribution of the PhPs excited only by the tip (as they show a $\lambda/2$ periodicity). The result shows a spiral with a fixed rotating center, as shown in Fig. S5c and S5d for the U- and L-RBs, respectively.

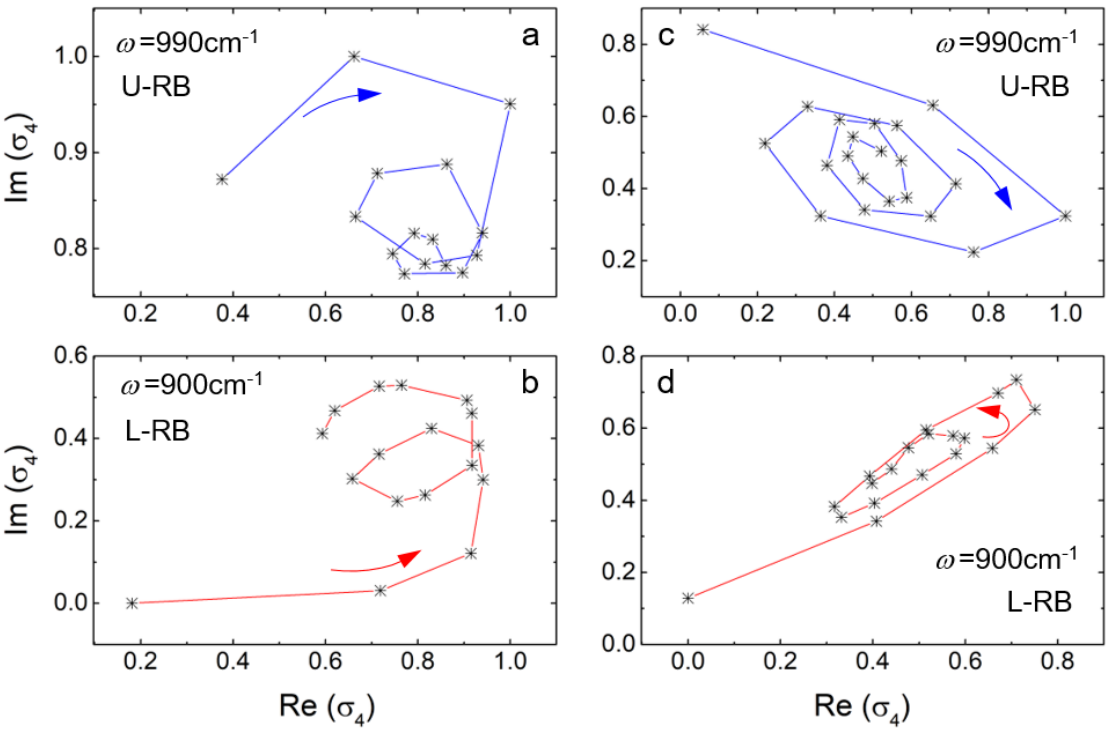

Figure S5. **a,b.** Imaginary and real parts of the complex-valued s-SNOM signal σ_4 extracted from Fig. 2b of the main text at ω= 990 cm^{-1} (U-RB), and ω= 900cm^{-1} (L-RB), respectively. **c,d.** Complex representation of the PhPs amplitude and phase profiles excited by the tip.

S.4. Group velocities of α-MoO_3 PhPs.

To extract the group velocity of α-MoO_3 PhPs, which is defined as $v_g = \frac{\partial\omega}{\partial k_i}$ (i= x, y), we take the derivative of the experimental dispersion curves $\omega(k_i)$ along the [100] and [001] directions shown in Fig. 2 of the main text. To do this, we first fit numerically the experimental data points of both RBs (red and blue symbols in Fig. S6a and S6b) by a generic potential function of the form $y = a \cdot x^b$ and finally perform a numerical derivative of the resulting curves (Fig. S6c, and S6d).

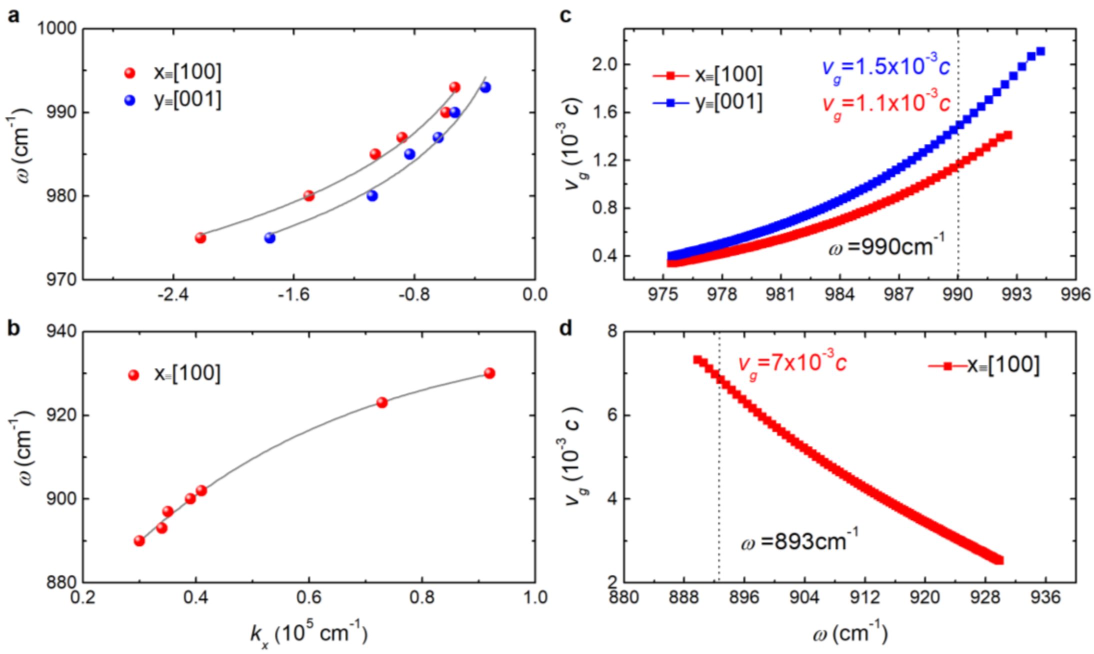

Figure S6. a,b. PhPs dispersion data points, $\omega(k_i)$, for the [100] (red symbols) and [001] (blue symbols) directions of α-MoO_3 crystal in the U- and L-RB, respectively. Gray lines are fits by a potential function. **c,d.** Group velocities, v_g, of PhPs in a 250-nm-thick MoO_3 flake (shown in Fig. 2b of the main text) for the [100] (red curve) and [001] (blue curve) directions of α-MoO_3 crystal in the U- and L-RB, respectively. The vertical dashed lines indicate the values (in light-speed units) obtained for $\omega = 990$ cm^{-1}, and 893 cm^{-1}.

S.5. Edge- and tip-launched PhPs in α-MoO_3.

Real-space s-SNOM polaritonic imaging relies on the capability of an AFM metallic tip to efficiently launch and detect polaritons. However, generally, apart from the tip, an inhomogeneity such as the sample edge can act as a polaritonic source[12,13]. When this happens, a simple way to distinguish between both contributions is to analyze the PhPs periodicity considering that a characteristic λ/2 periodicity due to interference is obtained when the tip is the source. Another difference is that the excitation of edge-launched polaritons depends strongly on the orientation of the incident field E_{inc} respect to the edge (when the incident field illuminates the edge longitudinally the excitation of PhPs is generally very low). This is clearly observed in the s-SNOM images of Fig. S7, where edge-launched α-MoO_3 PhPs are only observed when E_{inc} is perpendicular to the sample edge (Fig. S7b). This is more clearly observed in Figs. S7c, d where the Fourier transforms of the line profiles in Figs. S7a, b (white dashed lines) are plotted, respectively. The presence of double peaks in Fig. S7d confirms the presence of two different PhPs periodicities when E_{inc} is perpendicular to the sample edge, as in that case both the tip and the sample edge act as PhPs sources.

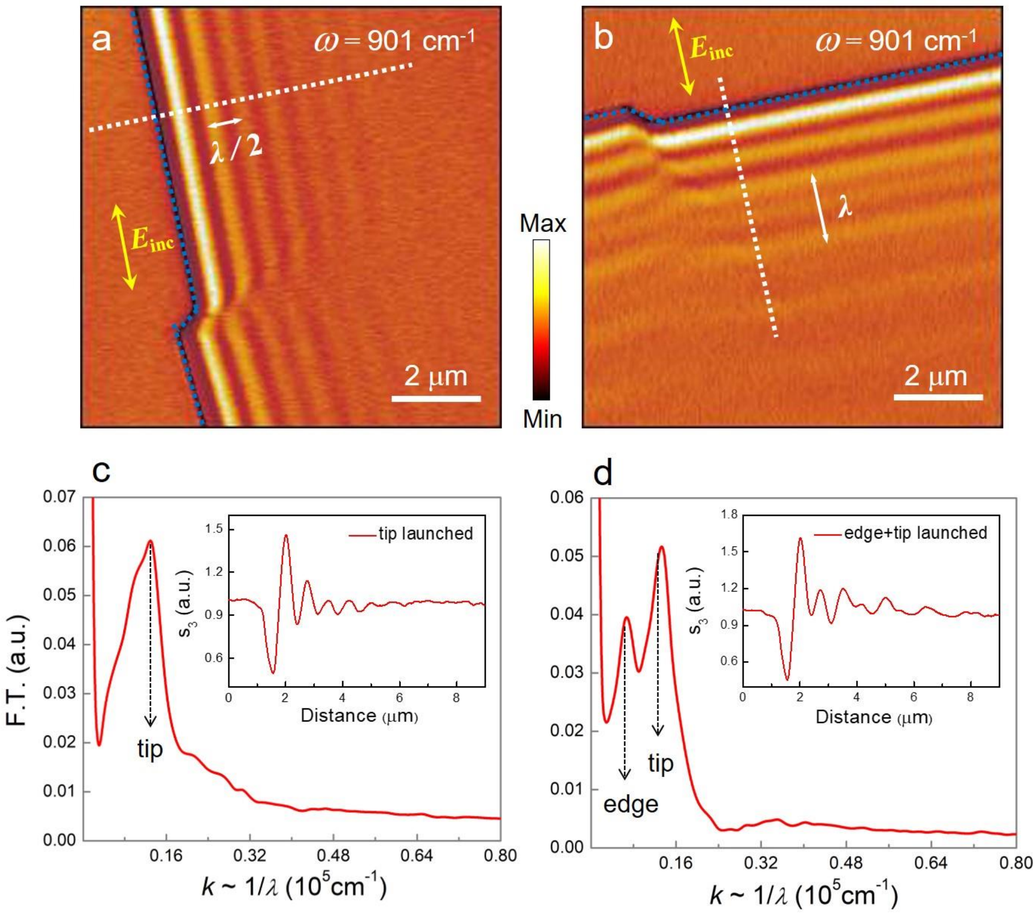

Figure S7. Edge and tip-launched PhPs in α-MoO_3. s-SNOM near-field amplitude image of an α-MoO_3 flake when the incident field is **a.** parallel and **b.** perpendicular to the sample edge. **c.**, **d.** Fourier transform of the line profiles along white dashed lines in **a.** and **b.**, respectively.

S.6. s-SNOM polariton interferometry in α-MoO_3.

For the interpretation of s-SNOM polariton interferometry images we have to consider that the component of the polariton momentum parallel to the reflecting material boundary, $\overrightarrow{k_{II}}$, has to be conserved (in case of an elastic reflection process[14]). For polariton back-reflection to the tip, this condition requires $\overrightarrow{k_{II}} = 0$. This back-reflection condition has to be satisfied for both in-plane isotropic and anisotropic polaritons (i.e. when the isofrequency curves are elliptic or hyperbolic), where momentum $\vec{k}$ and Poynting vector $\vec{S}$ are generally not collinear. The blue and red arrows in Fig. S8a illustrate the non-collinearity of wavevector and Poynting vector for a plane wave in the hyperbolic regime (L-RB) of α-MoO_3. Particularly, Fig. S8a,b illustrates how back-reflection of an in-plane hyperbolic polariton at an arbitrarily oriented flat boundary (mirror) can be understood and constructed geometrically with the help of the isofrequency curves. We consider a dipole-launched polariton of wavevector $\overrightarrow{k_\iota}$ and Poynting vector $\overrightarrow{S_\iota}$, which is propagating towards a mirror (illustrated in Fig. S8b, left). The back-reflection condition $\overrightarrow{k_{II}} = 0$ (see comment above) implies that the incident and reflected momenta $\overrightarrow{k_\iota}$ (blue arrow) and $\overrightarrow{k_r}$ (green arrow) are antiparallel and perpendicular to the mirror. However, the Poynting vector $\overrightarrow{S_\iota}$ (red arrow) is not parallel to $\overrightarrow{k_\iota}$, since $\overrightarrow{S_\iota}$ is normal to the hyperbolic isofrequency contour (Fig. S8a). For the same reason, $\overrightarrow{S_r}$ (black arrow) is not parallel to $\overrightarrow{k_r}$ but antiparallel to $\overrightarrow{S_\iota}$. Subsequently, not only the wavefronts but also the energy is back-reflected to the source. Interestingly, the energy flow is back-reflected although it is not normal to the mirror surface (anomalous reflection[15]).

We further illustrate the anomalous back-reflection in Fig. S8c, where the simulated near-field image of the α-MoO_3 disk is shown (see section S7-iii). The white dashed line in the figure indicates the direction of $\overrightarrow{k_\iota}$=-$\overrightarrow{k_r}$, which is perpendicular to the fringes of periodicity $\lambda/2 = \pi/k_i$. Note that the fringe periodicity is half the polariton wavelength in case the tip moves in the direction of $\vec{k}$, rather than in the direction of $\vec{S}$.

Importantly, all tip-launched polaritons (with all possible $\overrightarrow{k_\iota}$) can be back-reflected to the tip owing to the circular edge of the α-MoO_3 sample. For that reason, all possible $\overrightarrow{k_\iota}$ are probed when the whole disk is scanned, allowing for reconstructing the isofrequency contour in k-space via Fourier transform (FT) of the real space image. The FT of the simulated real-space s-SNOM image (Fig. S8c) is shown in Fig. S8d. A clear hyperbola is obtained, which exhibits an almost perfect matching to the hyperbolic contour obtained analytically by fitting the experiments (Fig. 3 of the main text).

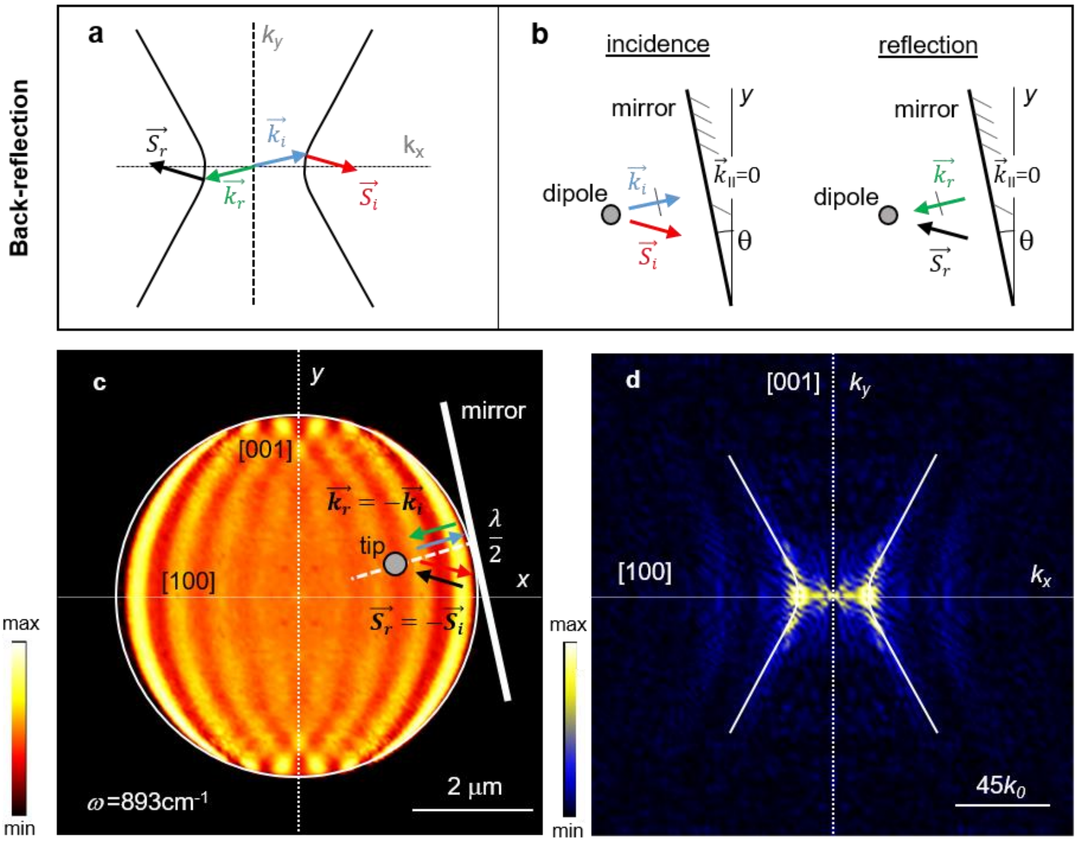

Figure S8. a. α-MoO_3 PhPs isofrequency contour at ω = 893 cm^{-1} (solid lines taken from Fig. 3d of the main text). Generally, $\vec{k}$ and $\vec{S}$ are non-collinear. **b.** Schematics of a wave back-reflection in an anisotropic medium. **c.** Calculated near-field amplitude image for an α-MoO_3 disk at ω = 893 cm^{-1}. The dashed white line indicates the direction of s-SNOM polariton interferometric measurement (perpendicular to the mirror indicated by a white line) that probes $\vec{k_i}$. **d.** Absolute value of the Fourier transform of the near fields in c, revealing a hyperbolic isofrequency contour. The solid lines show the PhPs isofrequency contours obtained by fitting Eq. 1 of the main text to the experimental s-SNOM interferometric images (Fig. 3 of the main text).

Reflection of in-plane isotropic/anisotropic PhPs at an interface with air

Fig. S9 shows numerically simulated $|E_z|$ created by a dipole source placed above the surface (z_d = 250 nm) of: i) a polaritonic in-plane hyperbolic material (α-MoO_3 at λ_0 = 11.2 μm) (Fig. S9a), and ii) a polaritonic in-plane isotropic material (h-BN at λ_0 = 6.8 μm) (Fig. S9b). The presence of a surface edge (marked by white dashed horizontal lines), produces interferometric fringes in both cases. However, whereas an isotropic near-field distribution is obtained for the isotropic case, a ray-like anisotropic interferometric pattern is observed in α-MoO_3, similar to reported results in Ref. 18.

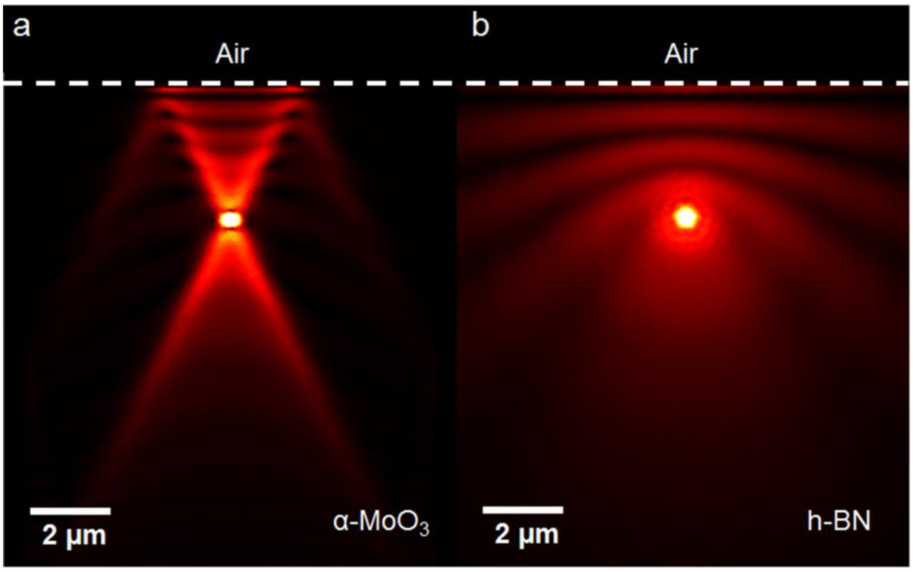

Figure S9. Reflection of in-plane isotropic and anisotropic PhPs. Simulated near-field spatial distribution $|E_z|$ of PhPs launched by a point dipole source on **a.** a hyperbolic α-MoO_3 surface, and **b**. an isotropic h-BN surface, in close proximity to an interface with air (dashed white line).

S.7. Derivation of the polariton dispersion along a thin anisotropic sheet.

Here we derive the dispersion relation for the polaritons in an anisotropic conducting sheet of zero thickness (occupying the plane $z = 0$) and placed between two dielectric half-spaces with the permittivities ε_1 (region "1", $z > 0$) and ε_2 (region "2", $z < 0$). We also derive the dependence of the polariton wavevector upon the thickness of a realistic slab.

i. Dispersion relation.

We assume that the conductivity tensor has the following form:

$$\hat{\sigma} = \begin{pmatrix} \sigma_{xx} & 0 \\ 0 & \sigma_{yy} \end{pmatrix}.$$

We represent the fields above ($z > 0$) and below ($z < 0$) the sheet using the following polarization basis vectors for the electric fields (in Dirac notations):

$$|s1{,}2\rangle = \frac{1}{k_t}\begin{pmatrix} -k_y \\ k_x \\ 0 \end{pmatrix} e^{ik_x x + ik_y y}, \qquad |p1{,}2\rangle = \frac{1}{k_t}\begin{pmatrix} k_x \\ k_y \\ \frac{k_t^2}{\mp k_{z1,2}} \end{pmatrix} e^{ik_x x + ik_y y} \qquad (1)$$

where k_x, k_y are in-plane wavevector components with $k_t^2 = k_x^2 + k_y^2$, and $k_{z1,2} = \sqrt{\varepsilon_{1,2} k_0^2 - k_x^2 - k_y^2}$ (with $k_0 = \frac{\omega}{c}$) is the out-of-plane wavevector component. $|s1{,}2\rangle$ and $|p1{,}2\rangle$ vectors correspond to the s- and p-polarization, respectively. The - (+) sign in the definition of the vector $|p1{,}2\rangle$ should be taken for the wave propagating along (opposite to) the z-axis, respectively, that is in the half-space 1 (half-space 2), respectively. We will also introduce the subvectors of the vectors given by Eq.(1), to represent the in-plane components of the fields:

$$|s\rangle = \frac{1}{k_t}\begin{pmatrix}-k_y\\ k_x\end{pmatrix} e^{ik_x x+ik_y y}, \qquad |p\rangle = \frac{1}{k_t}\begin{pmatrix}k_x\\ k_y\end{pmatrix} e^{ik_x x+ik_y y}. \quad (2)$$

The fields of the polariton in the upper and lower media can be compactly written as:

$$\vec{E}_1 = \sum_\beta a_\beta^+ |\beta 1\rangle\, e^{ik_{z1}z}, \qquad \vec{E}_2 = \sum_\beta a_\beta^- |\beta 2\rangle e^{-ik_{z2}z}, \, (3)$$

where β corresponds to the polarizations s, p, and a_β^+, a_β^- are the unknown amplitudes.

According to the Maxwell equations, the magnetic fields above and below the conducting sheet read as:

$$\vec{H}_1 = \vec{q}_1 \times \vec{E}_1, \quad \vec{H}_2 = \vec{q}_2 \times \vec{E}_2, \qquad (4)$$

where $\vec{q}_{1,2}$ is the normalized wavevector, $\vec{q}_{1,2} = \frac{\vec{k}_{1,2}}{k_0}$, $\vec{k}_{1,2} = (k_x, k_y, \pm k_{z1,2})$, where in the last expression, 1 and 2 correspond to the signs + and -, respectively. At the conducting sheet ($z = 0$) we have the following boundary conditions:

$$\vec{E}_{1t} = \vec{E}_{2t}, \qquad (5)$$

$$\vec{e_z} \times \left(\vec{H}_1 - \vec{H}_2\right) = 2\hat{\alpha}\vec{E}_{1t}. \qquad (6)$$

Where for convenience we have introduced the normalized conductivity tensor $\hat{\alpha} = \frac{2\pi}{c}\hat{\sigma}$. The subscript "t" in Eqs. (5,6) means the in-plane components. Using Eqs. (3,4), we can rewrite Eqs. (5,6) more explicitly:

$$\sum_\beta a_\beta^+ |\beta\rangle = \sum_\beta a_\beta^- |\beta\rangle, \qquad (7)$$

$$\sum_\beta \left(a_\beta^+\, \vec{e}_z \times \vec{q}_1 \times |\beta 1\rangle - a_\beta^-\, \vec{e}_z \times \vec{q}_2 \times |\beta 2\rangle\right) = \sum_\beta 2a_\beta^+ \hat{\alpha}|\beta\rangle. \qquad (8)$$

In order to simplify Eq. (8), let us notice that:

$$\vec{e}_z \times \vec{q}_1 \times |\beta 1\rangle = -Y_\beta^+ |\beta\rangle,$$

$$\vec{e}_z \times \vec{q}_2 \times |\beta 2\rangle = -Y_\beta^- |\beta\rangle. \quad (9)$$

with $Y_\beta^\pm$ being the admittances for s- and p-polarized waves (propagating along or opposite to) the z-axis:

$$Y_s^\pm = \pm q_{z1,2}, \;\; Y_p^\pm = \pm\frac{\varepsilon_{1,2}}{q_{z1,2}}. \;\; (10)$$

Then, taking into account Eq. (9), we can rewrite Eq. (8) as:

$$-\sum_\beta (a_\beta^+\, Y_\beta^+ |\beta\rangle - a_\beta^- Y_\beta^- |\beta\rangle) = 2\sum_\beta a_\beta^+ \hat{\alpha}\, |\beta\rangle. \qquad (11)$$

Multiplying the above line by $\langle\beta|$, in which the exponential should be complex conjugated (scalar products), we get:

$$-\sum_{\beta'}(a^+_{\beta'}Y^+_{\beta'}\langle\beta|\beta'\rangle - a^-_{\beta'}Y^-_{\beta'}\langle\beta|\beta'\rangle) = 2\sum_{\beta'}a^+_{\beta'}\langle\beta|\hat{\alpha}|\beta'\rangle. \quad (12)$$

Since $\langle\beta|\beta'\rangle = \delta_{\beta,\beta'}$ (Kronecker symbol), we have from Eq. (12):

$$-\left(a^+_\beta Y^+_\beta - a^-_\beta Y^-_\beta\right) = 2\sum_{\beta'}a^+_{\beta'}\langle\beta|\hat{\alpha}|\beta'\rangle. \qquad (13)$$

By projecting Eq. (7) to $\langle\beta|$, we have the equality of the amplitudes above and below the conducting sheet:

$$a^+_\beta = a^-_\beta. \qquad (14)$$

Using Eq. (14), we can reduce Eq. (13) to:

$$-a_\beta\left(Y^+_\beta - Y^-_\beta\right) = 2\sum_{\beta'}a_{\beta'}\langle\beta|\hat{\alpha}|\beta'\rangle = 2\sum_{\beta'}a_{\beta'}M_{\beta\beta'}, \qquad (15)$$

where we have removed the sign in the notation of a^+_β and a^-_β due to Eq. (14). Eq. (15) can be rewritten in the expanded form as:

$$\begin{cases}\left(M_{pp} + \frac{Y^+_p - Y^-_p}{2}\right)a_p + M_{ps}a_s = 0\\ \left(M_{ss} + \frac{Y^+_s - Y^-_s}{2}\right)a_s + M_{sp}a_p = 0\end{cases}, \qquad (16)$$

where for compactness we have introduced the following matrix $\widehat{M}$:

$$M_{\beta\beta'} = \langle\beta|\hat{\alpha}|\beta'\rangle. \qquad (17)$$

The solution of the system (16) has nontrivial values only when its determinant equals to zero:

$$\left(M_{pp} + \frac{Y^+_p - Y^-_p}{2}\right)\left(M_{ss} + \frac{Y^+_s - Y^-_s}{2}\right) - M_{sp}M_{ps} = 0. \qquad (18)$$

The latter equation constitutes the dispersion relation for polaritons. Let us find the explicit expression for the elements of the matrix $\widehat{M}$:

$$\langle p|\hat{\alpha}|p\rangle = \frac{1}{k_t^2}\begin{pmatrix}k_x & k_y\end{pmatrix}\begin{pmatrix}\alpha_{xx} & 0\\ 0 & \alpha_{yy}\end{pmatrix}\begin{pmatrix}k_x\\ k_y\end{pmatrix} = \frac{k_x^2\alpha_{xx} + k_y^2\alpha_{yy}}{k_t^2},$$

$$\langle p|\hat{\alpha}|s\rangle = \frac{1}{k_t^2}\begin{pmatrix}k_x & k_y\end{pmatrix}\begin{pmatrix}\alpha_{xx} & 0\\ 0 & \alpha_{yy}\end{pmatrix}\begin{pmatrix}-k_y\\ k_x\end{pmatrix} = \frac{-k_xk_y\alpha_{xx} + k_yk_x\alpha_{yy}}{k_t^2},$$

$$\langle s|\hat{\alpha}|p\rangle = \frac{1}{k_t^2}\begin{pmatrix}-k_y & k_x\end{pmatrix}\begin{pmatrix}\alpha_{xx} & 0\\ 0 & \alpha_{yy}\end{pmatrix}\begin{pmatrix}k_x\\ k_y\end{pmatrix} = \frac{-k_xk_y\alpha_{xx} + k_yk_x\alpha_{yy}}{k_t^2},$$

$$\langle s|\hat{\alpha}|s\rangle = \frac{1}{k_t^2}\begin{pmatrix}-k_y & k_x\end{pmatrix}\begin{pmatrix}\alpha_{xx} & 0\\ 0 & \alpha_{yy}\end{pmatrix}\begin{pmatrix}-k_y\\ k_x\end{pmatrix} = \frac{k_y^2\alpha_{xx} + k_x^2\alpha_{yy}}{k_t^2}.$$

From the above relations we see that the matrix $\widehat{M}$ is symmetric: $M_{sp} = M_{ps}$. Using the explicit form of the matrix elements, the dispersion relation transforms to:

$$\left[k_x^2\alpha_{xx} + k_y^2\alpha_{yy} + \frac{k_0 k_t^2}{2}\left(\frac{\varepsilon_1}{k_{z1}} + \frac{\varepsilon_2}{k_{z2}}\right)\right]\left[k_y^2\alpha_{xx} + k_x^2\alpha_{yy} + \frac{k_t^2}{2k_0}(k_{z1} + k_{z2})\right] -$$

$k_x^2 k_y^2 \left(\alpha_{xx} - \alpha_{yy}\right)^2 = 0$. (19)

Let us also rewrite Eq. (19) using normalized wavevector components (dividing it by k_0^4):

$$\left[q_x^2\alpha_{xx} + q_y^2\alpha_{yy} + \frac{q^2}{2}\left(\frac{\varepsilon_1}{q_{z1}} + \frac{\varepsilon_2}{q_{z2}}\right)\right]\left[q_y^2\alpha_{xx} + q_x^2\alpha_{yy} + \frac{q^2}{2}(q_{z1} + q_{z2})\right] -$$

$q_x^2 q_y^2 \left(\alpha_{xx} - \alpha_{yy}\right)^2 = 0$, (20)

where $q^2 = q_x^2 + q_y^2$. In case of the symmetric vacuum environment, i.e. when $\varepsilon_1 = \varepsilon_2 = 1$, after some straightforward algebra, Eq. (19) simplifies to (compare to Ref. 16):

$k_z k_0 \left(1 + \alpha_{xx}\alpha_{yy}\right) - \left(k_x^2\alpha_{xx} + k_y^2\alpha_{yy}\right) + k_0^2\left(\alpha_{xx} + \alpha_{yy}\right) = 0$, (21)

where $k_z = \sqrt{k_0^2 - k_x^2 - k_y^2}$.

ii. Thickness dependence of the polariton wavevector.

In order to obtain the dependence of the wavevector of the polaritons upon the thickness of the flake, d, we should use the relation between the normalized conductivity and the dielectric permittivity tensors:

$\hat{\sigma} = (c\mathrm{d}/2i\lambda_0)\hat{\varepsilon}$, or $\hat{\alpha} = (\pi\mathrm{d}/i\lambda_0)\hat{\varepsilon}$. (21)

Let us find the thickness-dependence of momentum in case of the polariton propagation along the x-axis. Assuming that $q_y = 0$ (and then $q = q_x$), Eq.(20) becomes a product of two factors. We will consider only the factor related to the TM polarization (since the TM modes have the largest momenta), and therefore the dispersion relation becomes:

$\alpha_{xx} + \frac{1}{2}\left(\frac{\varepsilon_1}{q_{z1}} + \frac{\varepsilon_2}{q_{z2}}\right) = 0.$ (22)

When the momentum of the polariton is significantly larger than the momentum of light in free space (*i.e.*, $q \gg 1$), we can set:

$q_{z1} \approx q_{z2} \approx iq_x.$ (23)

Then, from Eq. (22) we obtain:

$$q_x \approx \frac{\varepsilon_1+\varepsilon_2}{2} \cdot \frac{i}{\alpha_{xx}}. \qquad (24)$$

Substituting here α_{xx} expressed via ε_{xx} from Eq. (21), we finally arrive at the very simple expression:

$$k_x \approx -\frac{\varepsilon_1+\varepsilon_2}{\mathrm{d}\cdot\varepsilon_{xx}}. \quad (25)$$

Analogously, for the propagation in the x-direction, we get:

$$k_y \approx -\frac{\varepsilon_1+\varepsilon_2}{\mathrm{d}\cdot\varepsilon_{yy}}. \quad (26)$$

Notice that while in the elliptic case (in which $\mathrm{Re}(\varepsilon_{xx}), \mathrm{Re}(\varepsilon_{yy}) < 0$), the propagation along both x- an y-directions is permitted, in the hyperbolic case, in which the real parts of either ε_{xx} or ε_{yy} are positive, the propagation along one of the axes is not allowed.

iii. Numerical simulations of s-SNOM polariton interferometry images.

Based on a previous work[17], we theoretically model the s-SNOM polariton interferometry images of α-MoO_3 disks using full-wave simulations (with the help of COMSOL Multiphysics software). In the simulations the illuminated s-SNOM tip is approximated by a point dipole source located in close proximity to the surface of the disk (z_d=265 nm). The dipole is vertically (z-) oriented to take into account that the tip in the experiment is an elongated pyramid oriented perpendicular (vertical) to the sample and is illuminated by p-polarized light. The scattering s-SNOM signal from the tip can be approximated by the z-component of the electric field below the dipole (note that in-plane electric fields do not couple to the vertically oriented dipole). We thus model the near-field images by calculating the vertical component of the near field below the dipole (z_{mes}=20 nm with respect to the surface) as a function of dipole position, $|E_z(x,y)|$ above the α-MoO_3 disks. As described in section S7.1, the α-MoO_3 layer is approximated by a conducting sheet with zero thickness and anisotropic effective in-plane conductivity (metasurface). The conductivity values $\hat{\sigma}_{\mathrm{eff}}$ are obtained by fitting the Fourier transforms of the s-SNOM images by iso-frequency contours given by the dispersion relation of the anisotropic conducting sheet (ellipsoids in the U-RB and hyperbolas in the L-RB), as described in the main text of the manuscript.

Accordingly, the α-MoO_3 disk is numerically modelled as a slab (with nominal values of 144 nm in thickness and 6 μm in diameter extracted from topography measurements) with $\hat{\sigma}_{\mathrm{eff}}$ values corresponding to ε_{xx} = 2.6 and ε_{yy} = 3.7 for ω = 983 cm^{-1} (U-RB), and ε_{xx} = -6.4 and ε_{yy} = 1.7 for ω = 893 cm^{-1} (L-RB). For ε_{zz}, we set a value of 5 (-5) for the hyperbolic (elliptic) regime. Note that we chose the sign of ε_{zz} such as the PhPs phase velocity matches that of the experiment in each regime (negative in the U-RB and positive in the L-RB, see section S.3). The imaginary parts of the permittivities (not obtained from the fit) were adjusted to a value of 0.005 to obtain the best matching of the experimental images.

S.8. Notes for the fabrication of the α-MoO_3 disk

α-MoO_3 flakes were first isolated on a Si/SiO_2 (250 nm) substrate by mechanical exfoliation using blue Nitto tape (Nitto Denko Co., SPV 224P) from bulk MoO_3 crystals that were grown via chemical vapour deposition. The transferred flakes were inspected with an optical microscope and characterized via atomic force microscopy, allowing the selection of a large and homogeneous flake with the desired thickness. The selected flake was fabricated into a disk by using focused Ga-ion beam (FIB) milling in a FEI Helios 600 Nanolab dual beam system. We came up with a strategy to protect the surface of α-MoO_3 from implantation of Ga ions. The flake was fully covered by a thin diamond mask (500 nm thick). The diamond mask was cut from a bulk diamond film using the FIB. The mask was attached to the tungsten tip of an Omniprobe micromanipulator using an in-situ platinum deposition and physically placed on top of the flake. After the FIB etching of the flake and the mask in a ring-shaped pattern, the outer mask, still attached to the Omniprobe, was then lifted away and cut off with the ion beam. The Omniprobe was then reattached to the central diamond disk by a small electron beam platinum deposition, and it was also lifted away to give the disk-shaped flake separated from the bulk flake by a ring shaped channel. The original flake before fabrication and the fabricated disk present the unchanged polariton wavelength at same excitation frequency (Fig. S10a,b), which indicate that our strategy protects the α-MoO_3 from the Ga-ion damage during the fabrication process.

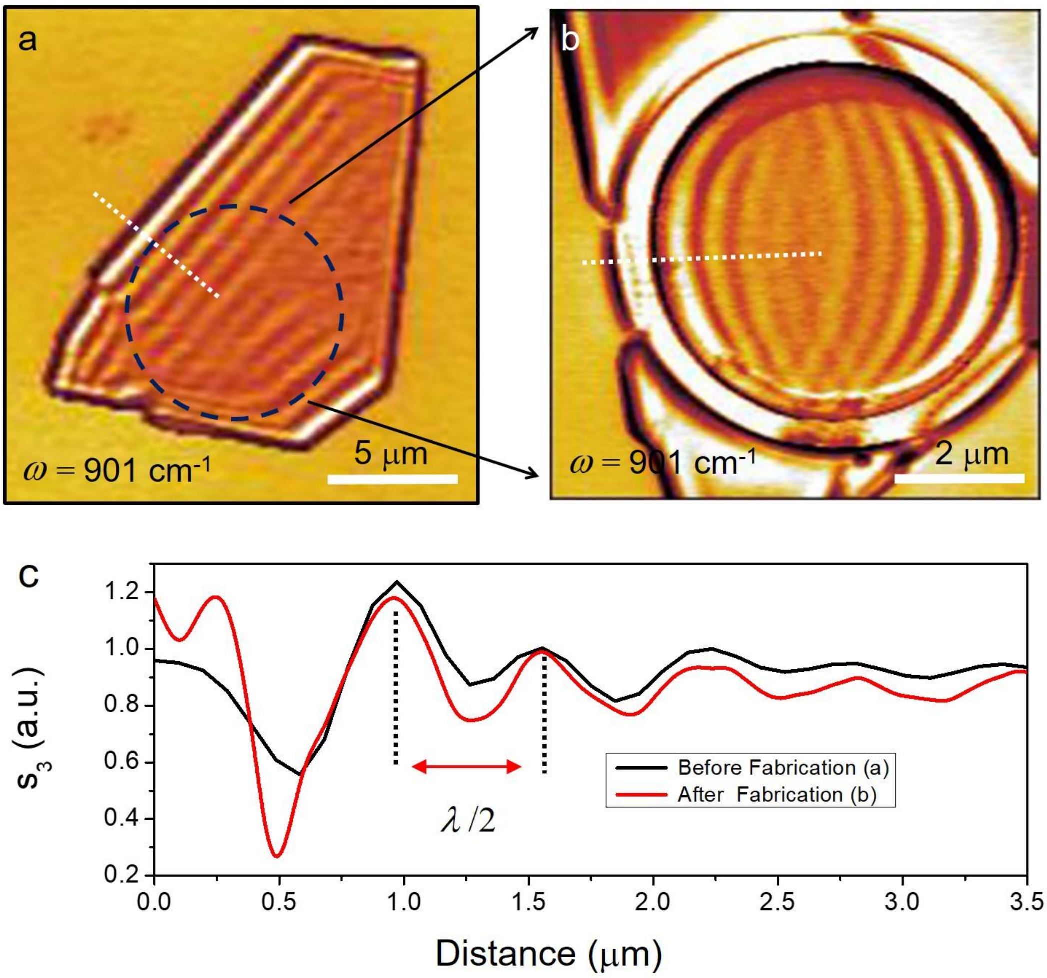

Figure S10. a. Near-field amplitude of an α-MoO_3 flake (before FIB fabrication) at ω = 896 cm^{-1}. The dashed circle indicates the area of fabricated nanodisk. **b.** Near-field amplitude image of fabricated α-MoO_3 disk at ω = 896 cm^{-1}. **c.** Line Profiles were extracted along the white dashed line in **a.** and **b.**, respectively. Similar polariton wavelength manifests the good feasibility of our protecting strategy.

S.9. Lifetimes of PhPs in α-MoO_3.

The lifetime of α-MoO_3 PhPs was calculated according to $\tau_x = L_x/v_g$, where the group velocity v_g was extracted as explained in section S.4., and the decay length L_x, by fitting a line profile of the PhPs near field (real part) along the [100] crystal direction. To simplify the fitting, we focused the analysis on edge-launched PhPs with a periodicity of λ (see section S.5) that propagate far from the edge (that way we exclude the contribution of tip-launched PhPs) (Fig. S11). For the fitting we assumed a sinusoidal signal (due to interference with the incident field), with a factor due to dissipation (described by an exponential decay of the PhPs field with distance x, see Eq. 27).

$$y = y_0 + Ae^{-\frac{x}{t_0}} \sin\left(\pi \frac{x - x_c}{w}\right), A > 0, w > 0, t_0 > 0 \qquad (27)$$

The experimental profiles Re(s_4) are also multiplied by a factor $\sqrt{x}$, in order to compensate the circular-wave geometrical spreading of the PhPs field[18,19,20] (see Fig. S12). In the fitting procedure, a Levenberg Marquardt iteration algorithm was applied, until full convergence was achieved ($\Delta\chi^2 \leq 10^{-9}$). From the fitting procedure we extract the PhPs propagation length as $L_x = t_0$.

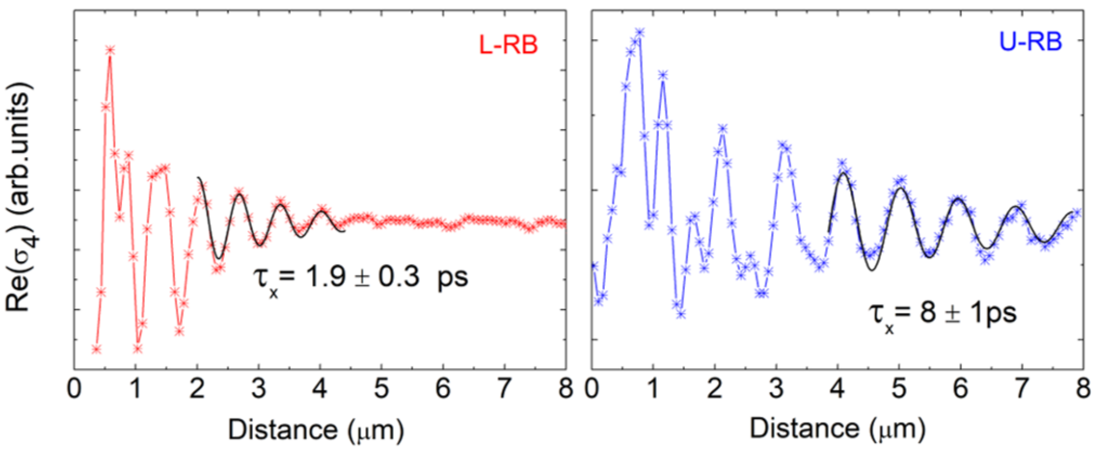

Figure S11. Near-field s-SNOM line traces (real part) along [100] direction of the flake shown in Fig. 2b of the main text in the hyperbolic (red crosses, ω = 930 cm^{-1}), and elliptic (blue crosses, ω = 990 cm^{-1}) regimes. Fits to a damped sine wave function are shown as black solid lines. The propagation lengths $L_x = t_0$ obtained were: t_0 = 1.45 ± 0.24 μm, and t_0 = 2.79 ± 0.37 μm in the L- and U-RB, respectively.

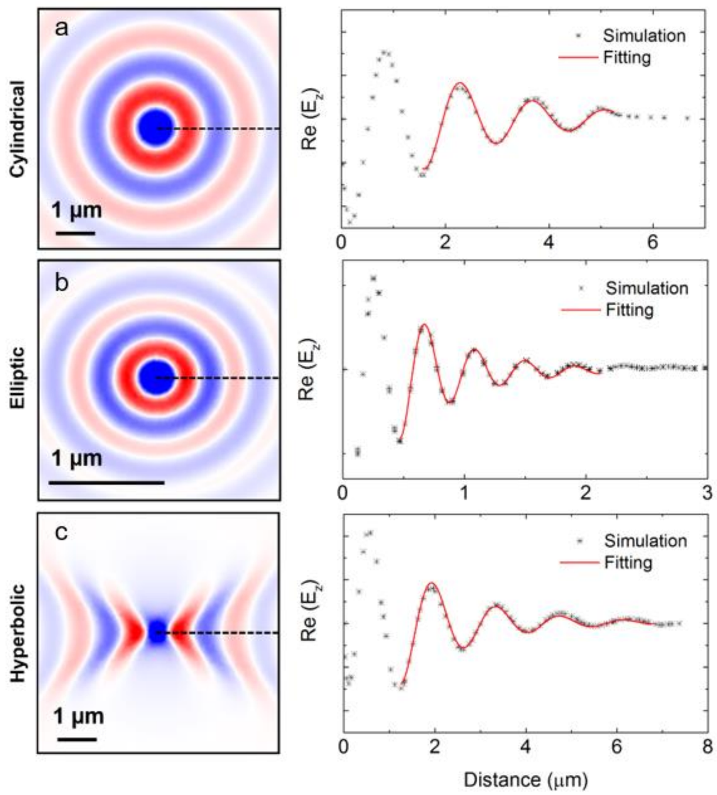

Figure S12. Numerical analysis of the geometrical spreading factor for circular, elliptic, and hyperbolic PhPs. **a.** Left: Field distribution of isotropically (radially) propagating PhPs. A point dipole is used to launch the PhPs. It is oriented perpendicular to an α-MoO_3 layer (with equal in-plane permittivities $\varepsilon_{xx} = \varepsilon_{yy}$ = -6.4, and an out-of-plane permittivity ε_{zz} = 5) and located 200 nm above its surface. Right: Field profile along the black dashed line shown in the left panel multiplied by $\sqrt{x}$ to compensate the geometric spreading, assuming that it can be well described by $\frac{1}{\sqrt{x}}$. By performing a fitting by an exponentially decaying sinusoidal function (red curve) an excellent agreement is obtained, thus validating the assumed geometric spreading factor $\frac{1}{\sqrt{x}}$. **b.** Left: Field distribution of elliptic PhPs on an α-MoO_3 layer (with in-plane permittivities ε_{xx} = 2.6, ε_{yy} = 3.7, and an out-of-plane permittivity ε_{zz} = -5). Right: Field profile along the black dashed line shown in the left panel multiplied by $\sqrt{x}$ to compensate the geometric spreading, assuming that it can be well described by $\frac{1}{\sqrt{x}}$. By performing a fitting by an exponentially decaying sinusoidal function (red curve) an excellent agreement is obtained, thus validating the assumed geometric spreading factor $\frac{1}{\sqrt{x}}$. **c.** Left: Field distribution of hyperbolic PhPs on an α-MoO_3 layer (with in-plane permittivities ε_{xx} = -6.4, ε_{yy} = 1.7, and an out-of-plane permittivity ε_{zz} = 5). Right: Field profile along the black dashed line shown in the left panel multiplied by $\sqrt{x}$ to compensate the geometric spreading, assuming that it can be well described by $\frac{1}{\sqrt{x}}$. By performing a fitting by an exponentially decaying sinusoidal function (red curve) an excellent agreement is obtained, thus validating the assumed geometric spreading factor $\frac{1}{\sqrt{x}}$.

S.10. Lifetime dependence of α-MoO_3 PhPs on the flake thickness.

To further corroborate the lifetime of α-MoO_3 PhPs in the U-RB shown in the main text (for a flake thickness d = 250 nm) and study it for various flake thicknesses d, we extracted the lifetime (at the same frequency ω = 990 cm^{-1}) for a flake with d = 160 nm (Fig. S13). The procedure was the same as for the 250-nm-thick flake, which is explained in section S.9. Particularly, the dispersion curves (Fig. S13a), and group velocities (v_g) (Fig. S13b) were extracted and calculated as explained in section S.4. The lifetime of α-MoO_3 PhPs was thus calculated according to $\tau_x = L_x/v_g$, where the decay length L_x (x indicates the [100] crystal direction) for each flake was extracted for ω = 990 cm^{-1} (Fig. S14). The resulting lifetime was 8 ± 1 ps (Fig. S14), and thus the same to that in the 250-nm-thick flake.

We also performed the same lifetime analysis in the L-RB for 4 different α-MoO_3 flakes with thicknesses d = 80, 100, 130, and 220 nm (Fig. S15, and S16). In all flakes, the extracted lifetimes were very similar (a deviation of 10%) with an average value of 2.1 ± 0.3 ps (Fig. S17). These results clearly indicate that the PhPs lifetime in α-MoO_3 can be considered as thickness-independent.

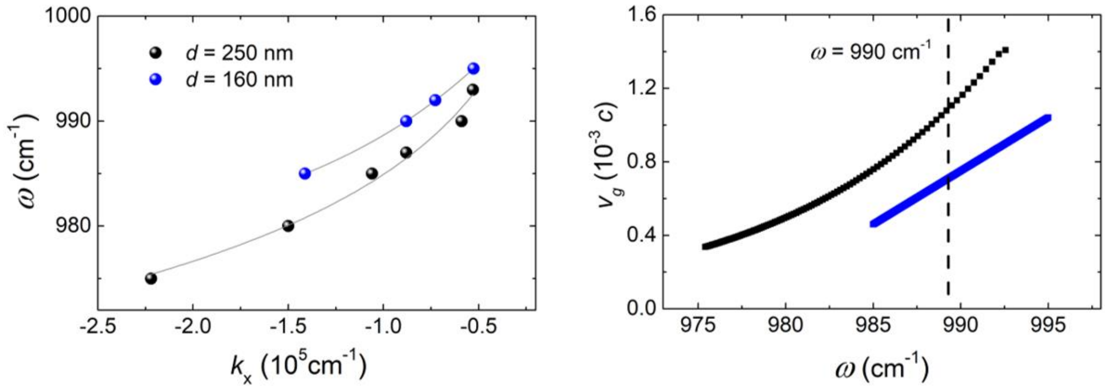

Figure S13. a. PhPs dispersion data points, $\omega(k_i)$, for the [100] direction of α-MoO_3 flakes with a thickness d = 160 (blue symbols), and 250 nm (black symbols). The gray lines are numerical fittings of the dispersions. **b.** Group velocities, v_g, of PhPs extracted by taking the derivative of the fittings in a (the color code is the same as in a). The vertical dashed line indicates the values (in light-speed units) obtained for ω = 990 cm^{-1}.

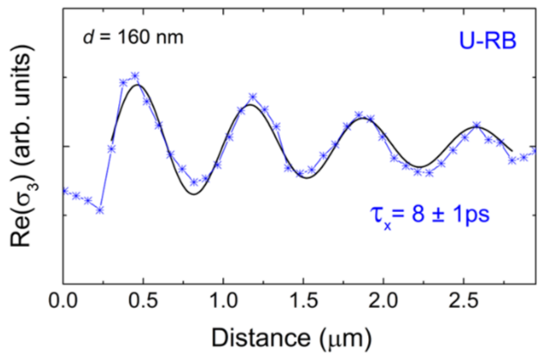

Figure S14. Near-field s-SNOM line traces along the [100] direction for a flake with thicknesses d = 160 nm (blue crosses, ω = 990 cm^{-1}). A fitting to a damped sine wave function is shown as a black solid line.

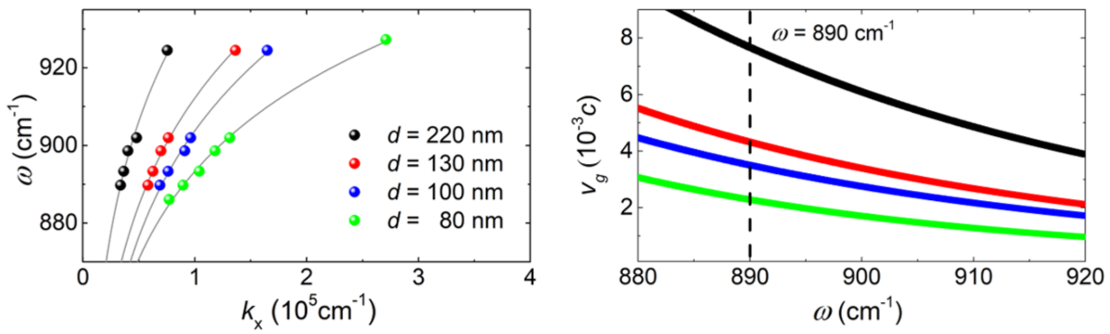

Figure S15. a. PhPs dispersion data points, $\omega(k_i)$, for the [100] direction of α-MoO_3 flakes with a thickness d = 80 (green symbols), 100 (blue symbols), 130 (red symbols), and 220 nm (black symbols). Gray lines are numerical fittings of the dispersions. **b.** Group velocities, v_g, of PhPs extracted by taking the derivative of the fittings in a (the color code is the same as in a). The vertical dashed line indicates the values (in light-speed units) obtained for ω = 890 cm^{-1}.

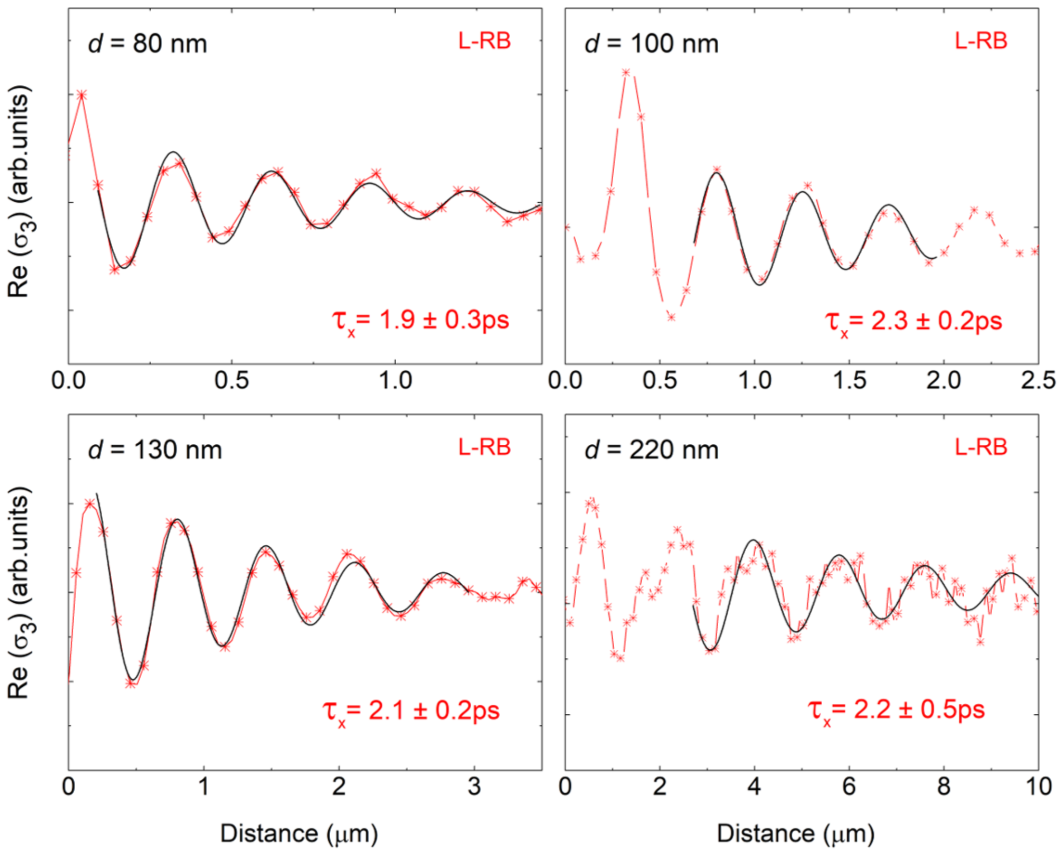

Figure S16. Near-field s-SNOM line traces along the [100] direction for 4 different flakes with thicknesses d = 80, 100, 130, and 220 nm (red crosses, ω = 890 cm^{-1}). Fits to a damped sine wave function are shown as black solid lines.

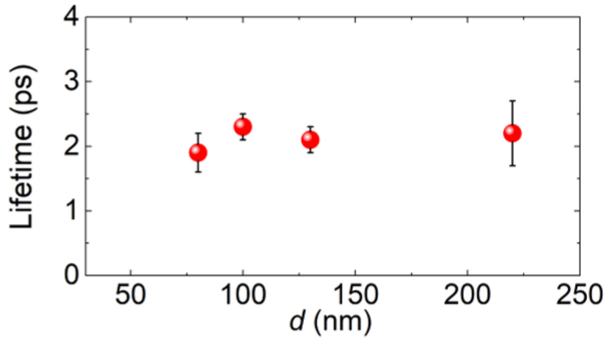

Figure S17. Lifetimes (symbols) extracted for 4 different flakes with thicknesses d = 80, 100, 130, and 220 nm.

Alternatively, the linewidth of the Raman peaks, which gives information of the crystal quality, can be related to the PhPs lifetimes, as recently reported in Ref. 6. We thus also analyzed the thickness dependence of the Raman signals at the 820 and 996 cm^{-1}, which correspond to the lattice vibrational modes that produce the optical phonons defining the L- and U-RBs, respectively (see Fig. S4d, e). Clearly, Fig. S18 shows a constant

FWHM as a function of flake thickness d for both Raman peaks. This result indicates that the PhPs lifetimes are expected to be thickness independent.

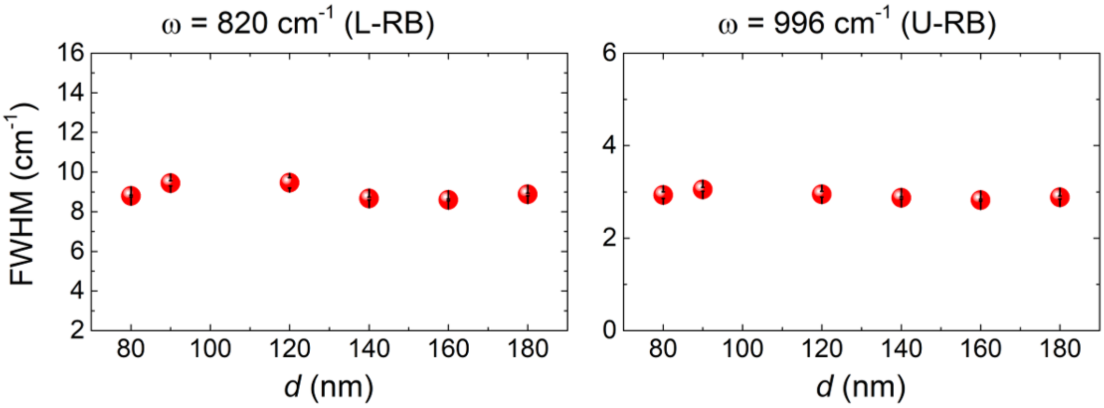

Figure S18. The red plots show the FWHM values of the peak extracted from the Raman spectra at 820 cm^{-1} (left) and 996 cm^{-1} (right) for 6 different flake thicknesses d = 80, 90, 120, 140, 160 and 180 nm.

S.11. Group velocity and lifetime of PhPs in an ultra-low-loss α-MoO_3 flake

Following the procedure explained in section S.4, we extracted the PhPs group velocity v_g in the U-RB for a 45-nm-thick α-MoO_3 flake. Due to the small thickness of the flake, the PhPs dispersion was very flat in this case (Fig. S19a), which thus yielded an extremely small group velocity $v_g = 3 \times 10^{-4}$ c at $\omega = 1002$ cm^{-1} (Fig. S19b). Using this group velocity v_g, the lifetime of α-MoO_3 PhPs was then calculated according to $\tau_x = L_x/v_g$, where the decay length L_x was obtained by fitting a line profile of the PhPs near field along the [100] crystal direction (Fig. S20a) and following the procedure described in section S.9 (Fig. S20b). The propagation length $L_x = t_0$ obtained was $t_0 = 2.1 \pm 0.4$ µm, which gives a record-high lifetime for ultra-confined polaritons of $\tau_p = 22 \pm 4$ ps. This lifetime is more than twice larger than those reported in section S.10 for flakes with d = 250 and 160 nm. This difference could be explained by the different frequencies ω used in each case (990 cm^{-1} and 1002 cm^{-1}) and thus of the different permittivities of α-MoO_3 in each case. Note that the infrared permittivity of α-MoO_3 is so far unknown beyond the values extracted in this work for 893 and 983 cm^{-1}.

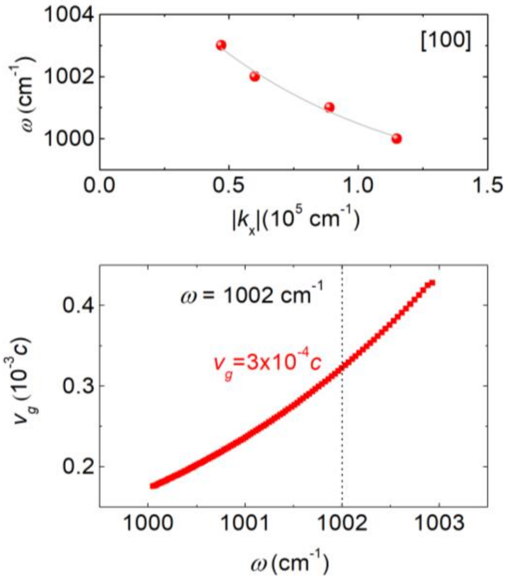

Figure S19. Upper panel: PhPs dispersion along the [100] direction in the U-RB. The gray line is a fitting. Lower panel: PhPs group velocity v_g for the [100] α-MoO_3 direction in the U-RB (obtained by taking the derivative of the gray line in the upper panel). The vertical line indicates the value (in light-speed units) obtained for ω = 1002 cm^{-1}.

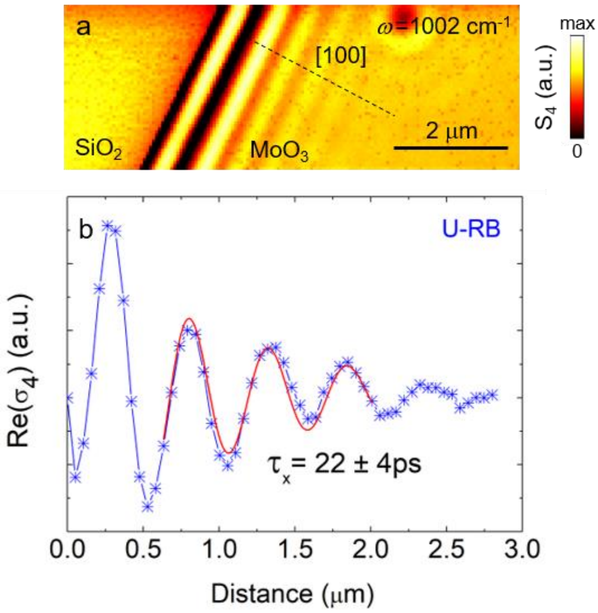

Figure S20. a. s-SNOM near-field amplitude image of an α-MoO_3 flake with a thickness d = 45 nm at illuminating frequencies ω = 1002 cm^{-1}. **b.** s-SNOM line traces along [100] direction (dashed line) of the flake shown in **a**. A fit to a damped sine wave function is shown as a red solid line.

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